

Computer Fundamentals, Memory and Number Systems

Chapter F9.5 — Back to a Real Foundation Layer, Because This Half Is Not Recall

Module	Foundation Layer — Module 9 — General Science & Computer Applications
Chapter	Chapter F9.5
Topic	Official syllabus head 6 — Basic Knowledge of Computer Applications
Version	v1.0
Compiled	16 August 2026
Exam stage	Recruitment Test (RT)

NOTE · THE SHAPE CHANGES HERE

F9.1 to F9.4 were written as a **recall layer**, because physics and chemistry are your undergraduate foundation. **Computer Applications is different**, and Blueprint §4.4 says so directly:

Number-system conversions are easy, but the C, HTML and operating-system items **need actual targeted preparation. Budget real hours there, not zero.**

So F9.5 to F9.7 are a **genuine Foundation layer** — built up from first principles, with worked conversions rather than stated results. The number systems in §5 are the part that rewards practice most, because they are **mechanical**: once you can do the grouping trick, you cannot get them wrong.

HIGH YIELD · THE 2023 COMPUTER BLOCK WAS TEN QUESTIONS

All ten 2023 Computer Applications items, and where each is covered:

Item	Chapter
Octal to binary conversion	F9.5 §5
Hexadecimal to decimal conversion	F9.5 §5
Program counter register	F9.5 §2
Real-time operating system	F9.6
Top-down languages	F9.6
C structure operator	F9.6
Hard disk error-check command	F9.6
IP address validity	F9.7
Mesh topology links	F9.7
HTML internal linking	F9.7

Three of the ten are in this chapter, and two of those three are pure conversion arithmetic — the cheapest marks in the entire paper, because the method is deterministic.

Table of Contents

0. How This Chapter Works

1. Session 1 — What a Computer Is ★★☆☆☆

1.1 The generations

2. Session 2 — The CPU and Its Registers ★★★★★

2.1 The three parts of the CPU

2.2 The machine cycle

2.3 The registers — the 2023 item

3. Session 3 — The Memory Hierarchy ★★★★★

3.1 The hierarchy, fastest first

3.2 RAM and ROM

3.3 Units of storage

4. Session 4 — Input and Output Devices ★★☆☆☆

5. Session 5 — Number Systems ★★★★★

5.1 The four systems

5.2 The grouping rule — the whole trick

5.3 Worked conversions

6. Session 6 — Codes and Logic Gates ★★☆☆☆

6.1 Character codes

6.2 Logic gates

7. Graded Practice — 30 Questions ★★★★★

7.1 Level 1 — Recognition

7.2 Level 2 — Understanding

7.3 Level 3 — Recruitment Test standard

8. Answers and Explanations

8.1 Level 1

8.2 Level 2

8.3 Level 3

9. One-Minute Revision

10. Five-Minute Wall Sheet

0. How This Chapter Works

STUDY SESSION · CHAPTER F9.5 — SIX SESSIONS

1. **Session 1** — What a computer is, and the generations (§1) ★★☆☆
2. **Session 2** — The CPU and its **registers** (§2) ★★★★★ **2023**
3. **Session 3** — The memory hierarchy (§3) ★★★★★
4. **Session 4** — Input and output devices (§4) ★★☆☆
5. **Session 5** — **Number systems and conversions** (§5) ★★★★★ **2023**
6. **Session 6** — Codes and logic gates (§6) ★★☆☆

Then 30 graded questions in §7. Budget **three days**, and spend at least a full session doing conversions until they are automatic.

1. Session 1 — What a Computer Is ★★☆☆☆

A **computer** is an electronic device that accepts **data** as input, **processes** it according to stored instructions, **stores** it, and produces **information** as output.

Data are raw facts; **information** is processed, meaningful data. The distinction is examined.

Characteristics: speed, **accuracy**, **diligence** (no fatigue), versatility, storage — and no intelligence of its own, which is the point of **GIGO: garbage in, garbage out**.

1.1 The generations

Generation	Technology	Examples
First	Vacuum tubes	ENIAC, UNIVAC
Second	Transistors	—
Third	Integrated circuits (IC)	—
Fourth	Microprocessors, VLSI	Personal computers
Fifth	Artificial intelligence, ULSI	—

MUST MASTER · THE GENERATION SEQUENCE, AND BABBAGE

Vacuum tube » transistor » integrated circuit » microprocessor » artificial intelligence.

Five stages, in that order, and any of them can be asked by number.

The two most-asked: **third generation = integrated circuit, fourth generation = microprocessor.**

Charles Babbage is called the **father of the computer**, for the Difference Engine and Analytical Engine. **Ada Lovelace** is regarded as the first programmer.

Types by capacity: supercomputer (India's include the PARAM series) » mainframe » mini » micro or personal computer. By data handling: analog, digital and hybrid.

CHECK YOURSELF · SESSION 1

1. Distinguish data from information
2. Name the technology of each of the five generations, in order
3. Who is the father of the computer? Who was the first programmer?
4. What does GIGO stand for, and what does it imply?

2. Session 2 — The CPU and Its Registers ★★★★★

2.1 The three parts of the CPU

Part	Function
ALU — Arithmetic and Logic Unit	Performs arithmetic and logical operations
CU — Control Unit	Directs and coordinates all operations; fetches and decodes instructions
Registers	Very small, very fast storage inside the CPU

The CPU is often called the **brain of the computer**. The ALU and CU together with registers form the processor.

2.2 The machine cycle

FETCH » DECODE » EXECUTE » STORE

The order is asked, and it is the only sensible one: you cannot decode an instruction before fetching it.

2.3 The registers — the 2023 item

MUST MASTER · WHAT EACH REGISTER HOLDS

This is a 2023 question, and the registers are only distinguishable by **what they hold**.

Register	What it holds
Program Counter (PC)	The address of the NEXT instruction to be executed
Instruction Register (IR)	The instruction currently being executed
Memory Address Register (MAR)	The address of the memory location to be accessed
Memory Buffer Register (MBR)	The data being transferred to or from memory
Accumulator (AC)	Intermediate arithmetic results

The distinction that decides the question: **the Program Counter holds an ADDRESS, and it is the address of the NEXT instruction — not the current one**. The register holding the current instruction is the **Instruction Register**.

Both halves are traps. An option saying the program counter holds the *current* instruction is wrong on both counts — wrong register and wrong kind of content.

Buses connect the parts: the **data bus** carries data, the **address bus** carries addresses, and the **control bus** carries control signals.

CHECK YOURSELF · SESSION 2

1. Name the three parts of the CPU and what each does
2. State the four steps of the machine cycle in order
3. What does the program counter hold? What does the instruction register hold?
4. Which register holds intermediate arithmetic results?
5. Name the three buses

3. Session 3 — The Memory Hierarchy ★★★★★

3.1 The hierarchy, fastest first

Registers » Cache » Main memory (RAM) » Secondary storage (hard disk, SSD, optical)

As you move down: **speed decreases, capacity increases, cost per byte decreases**. All three trends move together, which makes the hierarchy easy to reconstruct.

Cache sits **between the CPU and main memory** and is faster than RAM. It comes in levels — L1, L2, L3 — with L1 the smallest and fastest.

3.2 RAM and ROM

MUST MASTER · VOLATILE VERSUS NON-VOLATILE

	RAM	ROM
Full form	Random Access Memory	Read Only Memory
Volatility	VOLATILE — contents lost when power goes	NON-VOLATILE — contents retained
Access	Read and write	Read only
Purpose	Working memory for running programs	Stores the BIOS or firmware

RAM is volatile; ROM is non-volatile. That single pair answers a large share of memory questions, and the wrong option is almost always the reversal — "ROM is volatile memory".

Two types of RAM:

	SRAM (static)	DRAM (dynamic)
Refreshing	Not needed	Needs periodic refreshing
Speed and cost	Faster, costlier	Slower, cheaper
Used as	Cache	Main memory

Types of ROM: PROM — programmable once. **EPROM** — erasable by **ultraviolet light**. **EEPROM** — **electrically** erasable.

3.3 Units of storage

Unit	Size
Nibble	4 bits
Byte	8 bits
Kilobyte (KB)	1,024 bytes — that is 2^{10} , not 1,000
Megabyte, Gigabyte, Terabyte	Each 1,024 of the previous

Beyond terabyte: petabyte, exabyte, zettabyte, yottabyte.

CHECK YOURSELF · SESSION 3

1. List the memory hierarchy from fastest to slowest. What happens to cost and capacity?
2. Which of RAM and ROM is volatile? Which stores the BIOS?
3. SRAM or DRAM — which needs refreshing, and which is used as cache?
4. How is EPROM erased? How is EEPROM erased?
5. How many bits in a nibble? In a byte? How many bytes in a kilobyte?

4. Session 4 — Input and Output Devices ★★☆☆☆

Input devices	Output devices
Keyboard, mouse, scanner, joystick, light pen	Monitor, printer, plotter, speaker
OMR — optical mark reader	Impact printers: dot matrix , line
OCR — optical character reader	Non-impact: inkjet , laser , thermal
MICR — magnetic ink character recognition	—
Barcode reader, microphone, webcam	—

Both input and output: **touchscreen**, modem, network interface card.

FROM YOUR OWN WORLD · MICR IS ON EVERY CHEQUE YOU HAVE HANDLED

MICR — Magnetic Ink Character Recognition — is the technology used to read the code line printed at the bottom of a **bank cheque**. The characters are printed in magnetic ink so a machine can read them even if the cheque is stamped, signed over or slightly soiled.

You have handled tens of thousands of these. The exam asks "MICR is used mainly in..." and the answer is **cheque processing** — the one computer-applications question in this chapter you already know from the counter.

Note the contrast the options exploit: **OMR** reads pencil **marks** (answer sheets — including, probably, your own OMR sheet on RT day), **OCR** reads printed or written **characters** generally, and **MICR** reads **magnetic-ink** characters specifically.

CHECK YOURSELF · SESSION 4

1. Classify: scanner, plotter, touchscreen, MICR, laser printer
2. Distinguish OMR, OCR and MICR
3. Where is MICR used?
4. Which printers are impact printers?

5. Session 5 — Number Systems ★★★★★

Two 2023 questions came from this section, and it is the most mechanical topic in the entire syllabus. Learn the two grouping rules and the marks are guaranteed.

5.1 The four systems

System	Base	Digits used
Binary	2	0, 1
Octal	8	0 to 7
Decimal	10	0 to 9
Hexadecimal	16	0 to 9, then A B C D E F

MUST MASTER · THE HEXADECIMAL LETTERS STOP AT F

A = 10, B = 11, C = 12, D = 13, E = 14, F = 15.

F is 15, not 16, because the digits of base 16 run from 0 to 15 — there are sixteen of them, and the sixteenth is 15. And the letters run **A to F**, not A to G. Both errors are offered as options.

5.2 The grouping rule — the whole trick

MUST MASTER · THREE FOR OCTAL, FOUR FOR HEXADECIMAL

Because $8 = 2^3$ and $16 = 2^4$:

- Each **OCTAL** digit is exactly **THREE** binary digits
- Each **HEXADECIMAL** digit is exactly **FOUR** binary digits

So converting between binary and octal or hexadecimal requires **no arithmetic at all** — only grouping, **always from the right**, padding the leftmost group with zeros if needed.

Octal	Binary	Hex	Binary
0	000	0	0000
1	001	5	0101
2	010	9	1001
5	101	A	1010
7	111	D	1101
		F	1111

This is why octal and hexadecimal exist: they are compact ways of writing binary that a human can read, and the conversion is free.

5.3 Worked conversions

Octal to binary — the 2023 item. Replace each digit with its three bits:

725 in octal. $7 \gg 111 \cdot 2 \gg 010 \cdot 5 \gg 101$ Answer: **111 010 101**

Binary to octal. Group in threes from the right:

110110. $\gg 110 \mid 110 \gg 6 \mid 6 \gg$ **66 in octal**

Binary to hexadecimal. Group in fours from the right:

11011010. $\gg 1101 \mid 1010 \gg D \mid A \gg$ **DA in hexadecimal**

Hexadecimal to decimal — the other 2023 item. Multiply each digit by its positional power of 16:

2F. $= (2 \times 16) + (15 \times 1) = 32 + 15 =$ **47** **1A.** $= (1 \times 16) + (10 \times 1) = 16 + 10 =$ **26**

Binary to decimal. Add the positional weights where the bit is 1:

101101. = $32 + 8 + 4 + 1 = 45$ (weights from the right: 1, 2, 4, 8, 16, 32)

Decimal to binary. Divide repeatedly by 2 and read the remainders **upwards**:

25. $25 \div 2 = 12 \text{ r } 1$ · $12 \div 2 = 6 \text{ r } 0$ · $6 \div 2 = 3 \text{ r } 0$ · $3 \div 2 = 1 \text{ r } 1$ · $1 \div 2 = 0 \text{ r } 1$ Reading upwards: **11001**

MEMORY TECHNIQUE · CHECK EVERY CONVERSION IN FIVE SECONDS

Two habits that make errors nearly impossible:

1. **Group from the RIGHT, never the left.** The rightmost bit is the units bit, so grouping must start there. Grouping from the left is the single commonest mistake, and it gives an answer that looks plausible
2. **Sanity-check the magnitude.** A two-digit hexadecimal number lies between 0 and 255. So hex 2F cannot be 215 — that would need the first digit to be D. If your answer is out of range, you have multiplied by the wrong power

For decimal-to-binary, the reverse check takes five seconds: $11001 = 16 + 8 + 1 = 25$. Always do it.

CHECK YOURSELF · SESSION 5

1. Give the base and digit set of all four systems. What decimal value is F?
2. How many binary digits per octal digit? Per hexadecimal digit? Why?
3. Convert octal 356 to binary
4. Convert binary 10110110 to hexadecimal, and to octal
5. Convert hexadecimal 3C to decimal
6. Convert decimal 43 to binary, then check your answer

6. Session 6 — Codes and Logic Gates ★★☆☆☆

6.1 Character codes

Code	Bits	Characters
ASCII	7 bits standard; 8 bits extended	128; 256 extended
Unicode	16 bits and more	Covers most of the world's scripts
EBCDIC	8 bits	An IBM mainframe code
BCD	4 bits per digit	Binary coded decimal

6.2 Logic gates

Basic gates: **AND** (output 1 only if all inputs are 1) · **OR** (output 1 if any input is 1) · **NOT** (inverts).

Derived gates: **NAND**, **NOR**, **XOR**, **XNOR**.

MUST MASTER · NAND AND NOR ARE THE UNIVERSAL GATES

NAND and NOR are called universal gates, because either one alone can be used to build every other gate — AND, OR and NOT included.

The question is usually "which of the following is a universal gate", with AND, OR and NOT as distractors. **The basic gates are not universal; the two derived ones are.** That inversion is the whole point of the question.

CHECK YOURSELF · SESSION 6

1. How many bits does standard ASCII use, and how many characters does it encode?
2. Name the three basic and three derived gates
3. Which gates are universal, and what does universal mean?

7. Graded Practice — 30 Questions ★★★★★

7.1 Level 1 — Recognition

- Q1.** One byte is equal to (a) 4 bits (b) 8 bits (c) 16 bits (d) 1,024 bits
- Q2.** Which part of the central processing unit performs arithmetic and logical operations? (a) The control unit (b) A register (c) Cache memory (d) The arithmetic and logic unit
- Q3.** The register that holds the address of the **next** instruction to be executed is the (a) program counter (b) accumulator (c) instruction register (d) memory buffer register
- Q4.** RAM is (a) non-volatile and read-only (b) non-volatile and read/write (c) volatile and read/write (d) volatile and read-only
- Q5.** Which one of the following is **non-volatile** memory? (a) RAM (b) ROM (c) Cache (d) Register
- Q6.** The base of the hexadecimal number system is (a) 2 (b) 8 (c) 10 (d) 16
- Q7.** Charles Babbage is regarded as the father of the (a) computer (b) internet (c) transistor (d) integrated circuit
- Q8.** Which generation of computers used integrated circuits? (a) First (b) Second (c) Third (d) Fifth
- Q9.** MICR technology is used mainly in (a) supermarket billing (b) processing bank cheques (c) medical imaging (d) satellite imaging
- Q10.** One kilobyte is equal to (a) 1,000 bytes (b) 8 bytes (c) 512 bytes (d) 1,024 bytes

7.2 Level 2 — Understanding

- Q11.** The fastest of the following storage locations is (a) a register (b) cache memory (c) RAM (d) a hard disk
- Q12.** The octal number 725 expressed in binary is (a) 111 010 100 (b) 110 010 101 (c) 111 010 101 (d) 101 010 111
- Q13.** The hexadecimal number 2F expressed in decimal is (a) 31 (b) 47 (c) 45 (d) 215
- Q14.** To convert a binary number into octal, the bits are grouped from the right in sets of (a) two (b) eight (c) four (d) three
- Q15.** To convert a binary number into hexadecimal, the bits are grouped from the right in sets of (a) four (b) three (c) eight (d) two
- Q16.** The binary number 101101 expressed in decimal is (a) 41 (b) 43 (c) 45 (d) 47
- Q17.** Which one of the following is a **universal** logic gate? (a) AND (b) NAND (c) OR (d) NOT
- Q18.** DRAM differs from SRAM in that DRAM (a) is faster (b) is used as cache memory (c) is non-volatile (d) requires periodic refreshing

Q19. Standard ASCII uses (a) 7 bits (b) 4 bits (c) 16 bits (d) 32 bits

Q20. Which one of the following is **both** an input and an output device? (a) Scanner (b) Printer (c) Touchscreen (d) Keyboard

7.3 Level 3 — Recruitment Test standard

Q21. Which one of the following statements is **not** correct? (a) A register is faster than cache memory (b) ROM is volatile memory (c) The program counter holds the address of the next instruction to be executed (d) DRAM requires periodic refreshing

Q22. Consider the following statements:

1. Each octal digit corresponds to exactly three binary digits.
2. Each hexadecimal digit corresponds to exactly four binary digits.
3. The hexadecimal digit F represents decimal 16.

Which of the above statements are correct? (a) 1 and 3 only (b) 2 and 3 only (c) 1, 2 and 3 (d) 1 and 2 only

Q23. Match List-I with List-II and select the correct answer using the code given below:

List-I	List-II
A. Arithmetic and logic unit	1. Arithmetic and logical operations
B. Control unit	2. Directs and coordinates operations
C. Program counter	3. Address of the next instruction
D. Accumulator	4. Intermediate arithmetic results

(a) A-1 B-2 C-3 D-4 (b) A-2 B-1 C-4 D-3 (c) A-1 B-2 C-4 D-3 (d) A-3 B-4 C-1 D-2

Q24. The binary number 11011010 expressed in hexadecimal is (a) AD (b) BA (c) DA (d) D6

Q25. The decimal number 25 expressed in binary is (a) 10011 (b) 11001 (c) 10101 (d) 11011

Q26. Consider the following statements about computer memory:

1. Cache memory lies between the CPU and main memory.
2. ROM typically stores the BIOS or firmware.
3. Registers are located in secondary storage.

Which of the above statements are correct? (a) 1 and 3 only (b) 2 and 3 only (c) 1, 2 and 3 (d) 1 and 2 only

Q27. The correct sequence of the machine cycle is (a) fetch, decode, execute, store (b) decode, fetch, execute, store (c) fetch, execute, decode, store (d) store, fetch, decode, execute

Q28. Which one of the following statements is **not** correct? (a) EPROM can be erased using ultraviolet light (b) A nibble consists of four bits (c) The hexadecimal system uses the digits 0 to 9 and the letters A to G (d) SRAM is faster and costlier than DRAM

Q29. Which generation of computers is associated with the microprocessor? (a) Second (b) Fourth (c) First (d) Third

Q30. The hexadecimal number 1A expressed in decimal is (a) 110 (b) 20 (c) 24 (d) 26

8. Answers and Explanations

8.1 Level 1

Q	Ans	Explanation
1	b	8 bits. A nibble is 4 bits — option (a) is the nibble
2	d	The arithmetic and logic unit . The control unit directs and coordinates but does not compute
3	a	The program counter — an address , and of the next instruction. The instruction register holds the <i>current</i> instruction
4	c	Volatile and read/write. ROM is the non-volatile, read-only one
5	b	ROM. RAM, cache and registers are all volatile
6	d	16
7	a	The computer
8	c	Third. First was vacuum tubes, second transistors, fourth microprocessors
9	b	Bank cheques — magnetic ink character recognition reads the code line
10	d	1,024 bytes , which is 2^{10} . Option (a) is the decimal approximation and is the standard distractor

8.2 Level 2

Q	Ans	Explanation
11	a	A register , inside the CPU. The hierarchy is register » cache » RAM » disk
12	c	Each octal digit is three bits: 7 » 111, 2 » 010, 5 » 101, giving 111010101 . Option (a) mis-converts the 5, and (d) reverses the order
13	b	$(2 \times 16) + 15 = \mathbf{47}$. Option (a) treats F as 15 but forgets to multiply the 2 by 16; (d) is a magnitude error — a two-digit hex number cannot exceed 255 but 215 would need a leading D
14	d	Three , because $8 = 2^3$
15	a	Four , because $16 = 2^4$
16	c	$32 + 8 + 4 + 1 = \mathbf{45}$
17	b	NAND . NAND and NOR are the universal gates; AND, OR and NOT are the basic ones and are not universal
18	d	Requires periodic refreshing — that is what "dynamic" means. SRAM is the faster, costlier one used as cache
19	a	7 bits , encoding 128 characters. Extended ASCII uses 8 bits for 256
20	c	A touchscreen both displays and accepts input

8.3 Level 3

Q	Ans	Explanation
21	b	ROM is NON-volatile — that is its defining property. The other three statements are correct
22	d	1 and 2 only. Statement 3 is false: F is 15 , not 16, because base-16 digits run 0 to 15
23	a	A-1 B-2 C-3 D-4 — the four straightforward pairings
24	c	Group in fours from the right: 1101 = D , 1010 = A , giving DA . Option (a) reverses the two groups, which is what grouping from the left produces
25	b	$25 = 16 + 8 + 1 = \mathbf{11001}$. Check by adding the weights back — option (a) is the same digits reversed
26	d	1 and 2 only. Statement 3 is false — registers are inside the CPU , the fastest level of the hierarchy, not in secondary storage
27	a	Fetch, decode, execute, store. You cannot decode an instruction you have not fetched
28	c	Hexadecimal uses A to F , not A to G — there are only six letters, because 10 to 15 is six values
29	b	Fourth generation
30	d	$(1 \times 16) + 10 = \mathbf{26}$

THINK LIKE UPSC · THE FOUR SHAPES IN THIS CHAPTER

- Group from the RIGHT.** Q24's distractor (a) is exactly what you get by grouping from the left. This single habit protects both conversion questions
- The volatility reversal.** Q4, Q5 and Q21. **RAM volatile, ROM non-volatile** — state it as a pair every time
- The off-by-one in hex.** Q22 and Q28. **F = 15, letters A to F.** Both errors come from forgetting that base-16 digits start at zero
- Next versus current.** Q3 and Q21. The **program counter** holds the address of the **next** instruction; the **instruction register** holds the current one. Two registers, one word of difference

9. One-Minute Revision

ONE-MINUTE REVISION · CHAPTER F9.5

COMPUTER accepts data, processes, stores, outputs **information** · **data are raw, information is processed** · **GIGO** · diligence means no fatigue

GENERATIONS 1 **vacuum tube** · 2 **transistor** · 3 **integrated circuit** · 4 **microprocessor** · 5 **artificial intelligence** · **Babbage = father of the computer**, Ada Lovelace first programmer

CPU ALU arithmetic and logic · **CU** directs and coordinates · **registers** fastest storage · **machine cycle** **FETCH » DECODE » EXECUTE » STORE** · buses: data, address, control

REGISTERS **PROGRAM COUNTER = ADDRESS OF THE NEXT INSTRUCTION** · **instruction register = the CURRENT instruction** · **MAR** = address to access · **MBR** = data in transit · **accumulator = intermediate results**

HIERARCHY **register » cache » RAM » secondary** · going down, speed **falls**, capacity **rises**, cost per byte **falls** · cache sits **between CPU and main memory**, L1 fastest

RAM vs ROM **RAM volatile, read/write** · **ROM NON-volatile, read-only, holds the BIOS** · **SRAM** no refresh, faster, costlier, used as **cache** · **DRAM needs refreshing**, main memory · **PROM** once · **EPROM erased by ultraviolet** · **EEPROM electrically** erased

UNITS **nibble 4 bits** · **byte 8 bits** · **1 KB = 1,024 bytes** (2^{10})

DEVICES **OMR reads marks** · **OCR reads characters** · **MICR reads magnetic ink on CHEQUES** · **touchscreen is both input and output** · impact printers: dot matrix, line · non-impact: inkjet, laser

NUMBER SYSTEMS **binary 2** · **octal 8** · **decimal 10** · **hexadecimal 16, digits 0–9 then A–F** · **A=10 ... F=15**, letters stop at F

THE GROUPING RULE **octal = 3 bits per digit** ($8 = 2^3$) · **hexadecimal = 4 bits per digit** ($16 = 2^4$) · **ALWAYS GROUP FROM THE RIGHT**

WORKED octal 725 » **111 010 101** · binary 110110 » octal **66** · binary 11011010 » hex **DA** · hex 2F » **47** · hex 1A » **26** · binary 101101 » **45** · decimal 25 » **11001**

DECIMAL TO BINARY divide by 2 repeatedly, read remainders **upwards**

CODES **ASCII 7 bits, 128 characters**; extended 8 bits, 256 · Unicode covers world scripts

GATES basic AND, OR, NOT · derived NAND, NOR, XOR · **NAND and NOR are the UNIVERSAL gates**

10. Five-Minute Wall Sheet

FIVE-MINUTE WALL SHEET · CHAPTER F9.5

OCTAL = 3 BITS · HEX = 4 BITS · GROUP FROM THE RIGHT

A=10 B=11 C=12 D=13 E=14 F=15 — letters stop at F

HEX TO DECIMAL: × 16 THEN ADD — $2F = 32 + 15 = 47$ · $1A = 16 + 10 = 26$

BINARY WEIGHTS FROM THE RIGHT: 1 2 4 8 16 32 64 128

DECIMAL TO BINARY: DIVIDE BY 2, READ REMAINDERS UPWARD

RAM VOLATILE · ROM NON-VOLATILE (BIOS)

SRAM = CACHE, NO REFRESH · DRAM = MAIN MEMORY, NEEDS REFRESH

REGISTER » CACHE » RAM » DISK

PROGRAM COUNTER = NEXT INSTRUCTION'S ADDRESS · INSTRUCTION REGISTER = CURRENT INSTRUCTION

FETCH » DECODE » EXECUTE » STORE

NIBBLE 4 · BYTE 8 · KB 1,024

GEN 3 = IC · GEN 4 = MICROPROCESSOR

MICR = CHEQUES · OMR = MARKS · OCR = CHARACTERS

ASCII = 7 BITS = 128 CHARACTERS

NAND AND NOR ARE UNIVERSAL

ONE LINE FOR THE INTERVIEW Octal and hexadecimal exist because 8 and 16 are exact powers of two, so each digit maps onto exactly three or four bits — which lets a human read a binary value compactly without any arithmetic at all.

ACTION · NEXT IN MODULE 9

Chapter F9.6 — Operating Systems, Software and Programming, covering four more 2023 items: the **real-time operating system**, **top-down languages**, the **C structure operator** and the **hard disk error-check command**.

Before you move on: convert octal 461 to binary, binary 10011100 to hexadecimal, and hexadecimal 4D to decimal — then state what the program counter holds and which of RAM and ROM is volatile.